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*Technical Specification*

**LTE;**
**Evolved Universal Terrestrial Radio Access Network (E-UTRAN);**
**X2 signalling transport**
**(3GPP TS 36.422 version 8.3.0 Release 8)**






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# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

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Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

# 1 Scope

The present document specifies the standards for Signalling Transport to be used across X2 interface. X2 interface is a logical interface between eNBs. The present document describes how the X2-AP signalling messages are transported over X2.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] IETF RFC 2460: "Internet Protocol, Version 6 (IPv6) Specification".

[3] IETF RFC 791(September,1981): "Internet Protocol".

[4] IETF RFC 2474 (December 1998): "Definition of the Differentiated Services Field (DS Field) in the IPv4 and IPv6 Headers".

[5] IETF RFC 4960 (September 2007): "Stream Control Transmission Protocol".

# 3 Definitions, symbols and abbreviations

## 3.1 Definitions

For the purposes of the present document, the terms and definitions given in [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in [1].

**X2**: logical interface between two eNBs.

## 3.3 Abbreviations

For the purposes of the present document, the abbreviations given in [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in [1].

| | |
|---|---|
| eNB | E-UTRAN Node B |
| DiffServ | Differentiated Service |
| IP | Internet Protocol |
| PPP | Point to Point Protocol |
| SCTP | Stream Control Transmission Protocol |

# 4. X2 signalling bearer

## 4.1 Function and protocol stack

X2 signalling bearer provides the following functions:

- Provision of reliable transfer of X2-AP message over X2 interface.
- Provision of networking and routeing function
- Provision of redundancy in the signalling network
- Support for flow control and overload protection

The protocol stack for X2 Signalling Bearer is shown in figure 4.1 and details on each protocol are described in the following sections.

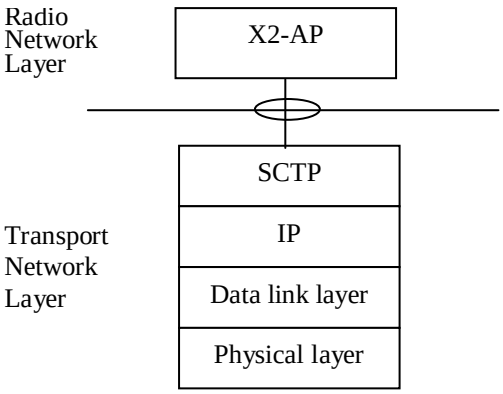



**Figure 4.1: X2 signalling bearer protocol stack**

The Transport Network Layer is based on IP transport, comprising SCTP on top of IP.

# 5 Data link layer

The support of any suitable Data Link Layer protocol, e.g. PPP, Ethernet, etc. , shall not be prevented.

# 6 IP layer

The eNB shall support IPv6 [2] and/or IPv4 [3].

The IP layer of X2 only supports point-to-point transmission for delivering X2-AP message.

The eNB shall support the Diffserv Code Point marking as described in [4].

# 7 Transport layer

SCTP [5] shall be supported as the transport layer of X2 signalling bearer.

SCTP refers to the Stream Control Transmission Protocol developed by the Sigtran working group of the IETF for the purpose of transporting various signalling protocols over IP network.

There shall be only one SCTP association established between one eNB pair.

An arbitrary eNB shall be able to initiate the INIT procedure towards another eNB for establishing the SCTP association.

Within the SCTP association established between one eNB pair;

- a single pair of stream identifiers shall be reserved for the sole use of X2AP elementary procedures that utilize non UE-associated signalling.
- a few pairs (i.e. more than one) of stream identifiers should be reserved for the sole use of X2AP elementary procedures that utilize UE-associated signalling.

Transport network redundancy may be achieved by SCTP multi-homing between two end-points, of which one or both is assigned with multiple IP addresses. SCTP end-points shall support a multi-homed remote SCTP end-point. For SCTP endpoint redundancy an INIT may be sent from either of the eNBs, at any time for an already established SCTP association, which shall be handled as defined in [5] in § 5.2.

The SCTP flow control may, using an implementation specific mechanism, initiate higher layer protocols to reduce the signalling traffic at the source and prioritise certain messages.

# 8 Service provided by the X2 signalling bearer

Editor"s Note: description of the necessary service to be provided by the S1-AP Signalling Bearer (if necessary).

# Annex A (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **TSG #** | **TSG Doc.** | **CR** | **Rev** | **Subject/Comment** | **Old** | **New** |
| 2007-11 | 38 | RP-070855 | | | specification presented to TSG-RAN for inforation and approval | | 1.0.0 |
| 2007-12 | 38 | | | | specification approved at TSG-RAN and placed under change control | 1.0.0 | 8.0.0 |
| 2008-03 | 39 | RP-080077 | 001 | - | Dedication of common streams over X2 | 8.0.0 | 8.1.0 |
| 2008-06 | 40 | RP-080301 | 003 | - | SCTP flow control and overload protection | 8.1.0 | 8.2.0 |
| 2008-09 | 41 | RP-080583 | 005 | 1 | Correct the wording of common and dedicated procedures | 8.2.0 | 8.3.0 |
| 2008-09 | 41 | RP-080583 | 006 | 2 | SCTP restart function for X2 interface | 8.2.0 | 8.3.0 |
| 2008-09 | 41 | RP-080583 | 007 | - | Clarification of SCTP Congestion Indication over X2 | 8.2.0 | 8.3.0 |
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# History

| Document history | | |
|---|---|---|
| V8.2.0 | November 2008 | Publication |
| V8.3.0 | November 2008 | Publication |
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